

Symbolic Derivation of GU-RSVP Stress-Energy Tensor and Christoffel Symbols

1 Action and Variation

The GU-RSVP action is:

$$S = \int d^4x \sqrt{-G} [\alpha \mathcal{R}_{\text{eff}}(G_{AB}, v^C, S) + \beta \nabla_A S \nabla^A \Phi - \gamma (\nabla_A v_B)(\nabla^A v^B) - V(\Phi, S)]$$

where $\mathcal{R}_{\text{eff}} = R + \lambda_1 \left(\frac{\delta S}{\delta \Phi}\right)^2 + \lambda_2 v^A v^B R_{AB}$, and $V(\Phi, S) = \mu \Phi^2 + \nu S^2 + \xi \Phi^2 S^2$.

Varying w.r.t. G_{AB} :

$$\frac{\delta S}{\delta G^{AB}} = \alpha \left(R_{AB} - \frac{1}{2} G_{AB} R \right) - T_{AB} = 0$$

The stress-energy tensor is:

$$T_{AB} = \beta (\nabla_A S \nabla_B \Phi + \nabla_B S \nabla_A \Phi - G_{AB} \nabla_C S \nabla^C \Phi) + \gamma (\nabla_A v_C \nabla_B v^C + \nabla_B v_C \nabla_A v^C - G_{AB} \nabla_D v_E \nabla^D v^E) - G_{AB}$$

2 Christoffel Symbols

For the effective metric G_{AB} :

$$\Gamma_{AB}^C = \frac{1}{2} G^{CD} (\partial_A G_{DB} + \partial_B G_{DA} - \partial_D G_{AB})$$

In a 4D projection (e.g., $G_{\mu\nu}$):

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2} G^{\rho\sigma} (\partial_\mu G_{\sigma\nu} + \partial_\nu G_{\sigma\mu} - \partial_\sigma G_{\mu\nu})$$

These are computed numerically in lattice simulations using finite differences.

3 Symbolic Implementation

Using SymPy:

```
from sympy import symbols, Function, diff, Matrix, IndexedBase
G, beta, gamma, mu, nu, xi = symbols('G beta gamma mu nu xi')
A, B, C, D = symbols('A B C D', cls=Idx)
Phi = Function('Phi')(x[C])
S = Function('S')(x[C])
v = IndexedBase('v')
V = mu * Phi**2 + nu * S**2 + xi * Phi**2 * S**2
T_AB = beta * (diff(S, x[A]) * diff(Phi, x[B]) + diff(S, x[B]) * diff(Phi, x[A]) -
               G[A,B] * diff(S, x[C]) * diff(Phi, x[C])) + \
gamma * (diff(v[C], x[A]) * diff(v[C], x[B]) + diff(v[C], x[B]) * diff(v[C], x[A]) -
         G[A,B] * diff(v[D], x[E]) * diff(v[E], x[D])) - G[A,B] * V
```